

Composing Piecewise and Nonlinear Functions

Evaluating a composition, writing the composed rule, and solving a composition equation

Directions

- $(f \circ g)(x)$ means $f(g(x))$: the inner function runs first.
- With a piecewise rule, the branch is chosen by the number going *into* that rule — check it every time you use the rule.
- Simplify composed rules fully, and give exact solutions.

1. Let $f(x) = x^2 + 1$ and $g(x) = 3x - 2$. Find $(f \circ g)(4)$.

Answer: _____

2. Let $p(x) = 2x + 5$ and $q(x) = x^2 - 3$. Write $(p \circ q)(x)$ as a simplified polynomial.

Answer: _____

3. Let $u(x) = x - 4$ and $v(x) = x^2$. Solve $(v \circ u)(x) = 49$.

Answer: _____

4. Let $g(x) = x^2 - 5$ and

$$f(x) = \begin{cases} 2x + 1, & x < 3 \\ x^2 - 4, & x \geq 3 \end{cases}$$

- (a) Find $(f \circ g)(4)$.

Answer: _____

- (b) Which rule for f did you use, and what comparison told you to use it?

Answer: _____

5. Let $f(x) = \frac{2}{x-1}$ and $g(x) = \frac{1}{x}$. Write $(f \circ g)(x)$ as a single simplified fraction.

Answer: _____

6. Let $f(x) = \sqrt{x+4}$ and $g(x) = 2x - 1$. Solve $(f \circ g)(x) = 5$.

Answer: _____

7. Let $f(x) = 3x - 2$ and

$$g(x) = \begin{cases} -x, & x \leq 0 \\ x^2, & x > 0 \end{cases}$$

- (a) Find $(f \circ g)(-5)$.

Answer: _____

- (b) Find $(g \circ f)(-5)$.

Answer: _____

8. Let $f(x) = x^2 - 6x$ and $g(x) = x + 3$. Write $(f \circ g)(x)$ as a simplified polynomial.

Answer: _____

9. Let $f(x) = x^2 + 2x$ and $g(x) = x - 3$. Solve $(f \circ g)(x) = 0$.

Answer: _____

10. Let

$$f(x) = \begin{cases} x + 7, & x < 2 \\ 3x, & x \geq 2 \end{cases}$$

(a) Find $(f \circ f)(-1)$.

Answer: _____

(b) Describe why the two passes through f do not use the same rule.

Answer: _____

- 11.** Let $f(x) = \frac{x}{x+1}$ and $g(x) = \frac{1}{x-2}$. Write $(f \circ g)(x)$ as a single simplified fraction.

Answer: _____

- 12.** Let $f(x) = \sqrt{x}$ and $g(x) = x^2 - 3x$.

(a) Solve $(f \circ g)(x) = 2$.

Answer: _____

- (b) Explain why neither of your answers is extraneous, even though one of them is negative.

Answer: _____